



# AI Risk Taxonomy for Financial Services

Complete risk register template and assessment methodology

**Generated:** May 11, 2026

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# 1. Executive Summary

AI systems in financial services introduce new and amplified risks across model, data, operational, and regulatory dimensions. This taxonomy provides a comprehensive framework for identifying, assessing, and managing these risks in a consistent manner across the enterprise.

# 2. Risk Categories

## Model Risk

**Model Performance Risk:** Model accuracy degrades or fails to perform as expected in production **Model Bias Risk:** Systematic favoritism toward certain groups or outcomes **Model Interpretability Risk:** Model decisions cannot be explained or justified **Model Stability Risk:** Model behaves unpredictably with small input changes

## Data Risk

**Data Quality Risk:** Training or production data is incomplete, inaccurate, or unrepresentative **Data Bias Risk:** Historical biases in data perpetuate or amplify systemic biases **Data Privacy Risk:** Personally identifiable information is exposed or misused **Data Governance Risk:** Unclear data ownership or inconsistent definitions

## Operational Risk

**Integration Risk:** AI system fails to integrate properly with existing systems **Governance Risk:** Insufficient controls and oversight of AI systems **Vendor Risk:** Third-party AI providers lack adequate controls **Change Management Risk:** Organization unable to adopt and use AI systems effectively

## Regulatory & Compliance Risk

**Fair Lending Risk:** AI systems discriminate based on protected characteristics **Consumer Protection Risk:** AI system fails to protect consumer interests **Explainability Risk:** Regulators cannot validate AI system decisions **Regulatory Change Risk:** New regulations make existing systems non-compliant

# 3. Risk Assessment Methodology

## Risk Scoring Matrix

Risk Level = Likelihood × Impact

| Rating       | Likelihood                      | Impact to Business                         |
|--------------|---------------------------------|--------------------------------------------|
| 1 - Low      | <10% probability in 12 months   | <\$100K loss / <1% operational impact      |
| 2 - Medium   | 10-50% probability in 12 months | \$100K-\$1M loss / 1-5% operational impact |
| 3 - High     | 50-90% probability in 12 months | \$1M-\$10M loss / 5-20% operational impact |
| 4 - Critical | >90% probability in 12 months   | >\$10M loss / >20% operational impact      |

## 4. AI Risk Register Template

Each AI system should have a documented risk register with minimum the following entries:

| Risk ID | Risk Category     | Description                        | Current State | Likelihood | Impact   | Overall Risk | Owner          | Mitigation                  |
|---------|-------------------|------------------------------------|---------------|------------|----------|--------------|----------------|-----------------------------|
| AIR-001 | Model Performance | Accuracy decline in validation set | Green         | Low        | Medium   | Medium       | Data Lead      | Weekly monitoring           |
| AIR-002 | Data Bias         | Historical bias in training data   | Amber         | Medium     | High     | High         | Analytics Lead | Synthetic data augmentation |
| AIR-003 | Regulatory        | New fair lending requirements      | Amber         | High       | Critical | Critical     | Compliance     | Impact assessment           |

## 5. Risk Indicators & Triggers

### Model Performance Indicators

• Accuracy drops >5% from baseline • Precision or recall declines >10% • F1 score below minimum threshold • Error rates exceed SLA • Unexpected patterns in residuals

### Data Quality Indicators

• Missing value rates >20% • Outlier prevalence increases >2x • Feature distributions shift >10% • Data freshness violations • Unexplained data changes

## 6. Mitigation Strategies

### Model Risk Mitigation

- Regular model validation and backtesting
- Bias testing on protected attributes
- Model explainability assessments
- Gradual rollout with monitoring
- Regular retraining schedules

### Data Risk Mitigation

- Data quality frameworks and SLAs
- Data audits for bias and representation
- Privacy-preserving techniques (anonymization, differential privacy)
- Data governance policies and metadata

■ Regular data lineage documentation

## 7. Regulatory Alignment

This taxonomy aligns with regulatory guidance from: • Federal Reserve SR 11-7 (Model Risk Management) • OCC Bulletin 2020-4 (AI/ML Governance) • FDIC Guidelines on Model Risk Management • Interagency Guidance on Fairness in Credit

## Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

### We're here to help.

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